REB FORM NO.-455(SUBSTATION PERIODIC INSPECTION AND MAINTENANCE FORM)

PBS :	Substation:	Capacity: =
Date of Commission:	Date of Augmentation:	Date of Maintenance:

PUT TICK (√) MARK BY ITEM INSPECTED OR MAINTAINED

I. POWER TRANSFORMER

ITEMS CHECKED		REMARKS
A. Manufacturer		
B. Sl.No		
C. Bushing chipped/broken/cracked/**Good**		
D. Record level oil		
E. Temperature gauge (winding)	Present(^{o}C)	
	Max(^{o}C)	
F. Temperature gauge (Oil)	present(^{o}C)	
	max(^{o}C)	
G. Record gas pressure (kg/cm^2 /lbs/in^2)		
H. Pressure release device operation		
I. Auxiliary fan operation		
J. General conditions of foundation		
K. Radiator, valve, plug etc.		
L. Conservator's tank, regulator, breather etc.		
M. Dielectric strength of oil (kv)		
N. Connection loose or corroded		
O. 240 Volt power supply & wiring		
P. % impedance		
Q. Tap changer position		
R. Vector group		

S. Megger results (GΩ):-

H1→x1		H1→x2		H1→x3	
H2→x1		H2→x2		H2→x3	
H3→x1		H3→x2		H3→x3	
H1→G		H2→G		H3→G	
x1→G		x2→G		x3→G	
H1→ H2		H2→ H3		H3→ H1	
x1→x2		x2→x3		x3→x1	

AGM(O&M)	Assistant Engineer	DGM(Technical)	Executive Engineer
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I. POWER TRANSFORMER

ITEMS CHECKED↓ REMARKS→		R- Φ		Y- Φ		B- Φ		Spare	
A. Manufacturer	1								
	2								
B. Sl.No.(as per previous connection)	1								
	2								
C. Sl.No.(as per new connection)	1								
	2								
Item checked for the Phases of									
D. Bushing chipped/broken/cracked									
E. Record level oil									
F. Winding temperature (present °C)									
(max °C)									
G. Oil/liquid temperature (present)									
(max)									
H. Record gas pressure (kg/cm^2 /lbs/in^2)									
I. Pressure release device operation									
J. Auxiliary fan operation									
K. General conditions & foundation									
L Leakage of radiator, valve, plug etc.									
M. Dielectric strength of oil (Kv)									
N. Connection loose or corroded									
O. 240 Volt power supply & wiring									
P. % impedance									
Q. Tap changer position									
R. Cooling Type									

S. Megger results(GΩ):

Φ R1:-	HT→LT =		HT→G =		LT→G =		HT→HT/LT→LT =
Φ R2:-	HT→LT =		HT→G =		LT→G =		HT→HT/LT→LT =
Φ Y1:-	HT→LT =		HT→G =		LT→G =		HT→HT/LT→LT =
Φ Y2:-	HT→LT =		HT→G =		LT→G =		HT→HT/LT→LT =
Φ B1:-	HT→LT =		HT→G =		LT→G =		HT→HT/LT→LT =
Φ B2:-	HT→LT =		HT→G =		LT→G =		HT→HT/LT→LT =

S .Bushing as viewed from HT (33Kv) side and LT (11Kv) side are as follows :- ◙ H2 ◙ H1 and ◙ X2 ◙ X1

T. Sl.No.(as installed, 33Kv→11 Kv):

U. **Φ** Identification:-

RΦ:- YΦ→H1, YΦ→H2, X2 → rΦ, X1 is grounded
YΦ:- BΦ→H1, BΦ→H2, X2 → yΦ, X1 is grounded
BΦ:- BΦ→H1, RΦ→H2, X2 → bΦ, X1 is grounded

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II. VOLTAGE REGULATOR

ITEMS CHECKED↓ REMARKS→	R-Phase	Y-Phase	B-Phase
A Manufacturer			
B. Sl.No.			
C. Control type			
D. Rating(KVA, Ampere)			
E. Bushing contamination			
F. Bushing chipped, broken or cracked			
G. Record level of oil			
H. Connection loose or corroded			
I. Oil tank damaged or leaking			
J Dielectric strength of oil(Kv)			
K. Operation Counter reading			
L. Megger result RΦ,(GΩ):- S/L/SL = S/L/SL/BDY =			
M. Megger result YΦ,(GΩ):- S/L/SL = S/L/SL/BDY =			
N Megger result BΦ,(GΩ):- S/L/SL = S/L/SL/BDY =			
O. Control panel settings:-Voltage level = , Bandwidth = , Time delay =			

Note:-

III. AUTOMATIC CIRCUIT RECLOSER

Items Checked↓ Remarks→	33 Kv Source	11 Kv Feeders					
		1	2	3	4	5	6
A. Manufacturer							
B. Control type							
C. Sl.No.							
D. Rating (Amps)							
E. Curve setting phase							
F. Curve setting ground							
G. Bushings							
H. Tank/body							
I. Operation Counter							
J. Op. Counter reading							
K. Mis-operation detect							
L. Moisture indication							
M. Misalignment/arcing							
N. Oil level indicator							
O. SF6 pressure (kPag)							
P. Oil strength (Kv)							
Q. Oil valve/plug/gasket							
R. Frame/structure							
S. Servicing							
T. Oil changed							
U. SF6 gas added							
V. Control’s function							

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IV . AUXILIARY EQUIPMENT:-

A. METER CABINET & CONTROL PANELS

Items Checked↓ Remarks→ 0	Remark	ITEMS	Remark
A. Wiring/Lighting/terminal blocks/switches		B. Tightness of cover& gaskets	
C. Meters type –			

B. CURRENT & POTENTIAL TRANSFORMER

Items Checked↓ Remarks→	Current Transformer		Potential Transformer	
	R-Phase	B- Phase	R-Y	B-Y
A. Manufacturer				
B. Sl. No.				
C. Ratio (Put √ for ratio used)				
D. Bushing contaminated				
E. Bushing broken or cracked				
F. Connections loose corroded				
G Tank/body damaged or leak				
H. Oil level /Strength (Kv)				
I. Megger result(GΩ):-				

V. SURGE ARRESTERS (Power Transformer)

Items Checked↓ Remarks→	Red Phase		Yellow Phase		Blue Phase		Spare
	33 Kv	11 Kv	33 Kv	11Kv	33 Kv	11 Kv	-
A. Manufacturer							
B. Sl. No.							
C. Rating (Kv)							
D. Megger result (GΩ)							
E. Porcelain broken/cracked							

VI ISOLATION SWITCHES (199 & 166)

Items Check↓ Rem→	199	166	Items Checked↓ Remarks→	199	166
A. Liver/Handle			E. Connection loose or corroded		
B. Blade, contacts			F. Insulator contaminated, chipped.		
C. Smooth operation			G. Base mounting, linkage& bearing		
D. Ground connection			H. Remarks		

VII ISOLATION SWITCHES (299 & 266)

Items Check↓ Rem→	299	266	Items Checked↓ Remarks→	299	266
A. Liver/Handle			E. Connection loose or corroded		
B. Blade, contacts			F. Insulator contaminated, chipped.		
C. Smooth operation			G. Base mounting, linkage& bearing		
D. Ground connection			H. Remarks		

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VIII 33 KV FUSES

Items Ch↓Rem→	R-Φ	Y-Φ	B-Φ	Items Check↓Rem→	RΦ	YΦ	BΦ
A. Manufacturer				E . Base & Mounting			
B. Rating (Amps)				F. Smooth Operation			
C. Fuse link/barrel				G. Alignment			
D. Contacts				H. Connection			

IX INSULATORS

Items Checked↓Rem→		Items Checked↓ Remarks→		Items Checked↓Rem→	
A.Porcelain broken/crack		B.Metal parts/base/mounting		C. Washing & cleaning	

X. PROTECTIVE LIGHTING

Items Ch.	Remark	Items Checked	Remark	Items Checked	Remark
Glassware		Photo Electric Cell Assembly		Conduit/fittings/fixtures	

XI. STRUCTURES (STEEL /WOODEN / SPC& STEEL/SPC & WODDEN)

ITEMS (Steel Structure)	Remarks	ITEMS (Wooden pole)	Remarks
A. Painting performed		A. Condition of poles	
B. Rust removed		B. Condition of Cross arm	
C. Loose broken & missing		C. Condition of guy/anchor	

XII. PAD & FOUNDATION

ITEMS	Remarks	ITEMS	Remarks
A. Steel Structure foundation		B. X-former pad	
C. V.R pad		D. OCR/ACR pad	

XIII. SUBSTATION YARD

ITEMS	Remarks	ITEMS	Remark
A. Fence/painting/intactness		B. Leveling /Slope/road	
C. Gate & its operation		D. Locks & keys	
E. Weed/flooding		F. Warning signs	

XIV. GROUNDING SYSTEM

ITEMS	Remarks	ITEMS	Remarks
A. Connections loose or corroded		B. Grounding resistance measurement	

XV. TOOLS AND EQUIPMENT

ITEMS	Remarks	ITEMS	Remarks
A. Quantity & condition checked		B. Storage area cleaned	

XVI. SPARE PARTS

ITEMS	Remarks	ITEMS	Remarks
A. Verify quantities		B. Stored condition	

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XVII. REMARKS / RECOMMENDATION:-

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